

Bias Variance Decomposition

Till now, all the learning algorithms we saw seek to find a linear predictor y that minimizes the some error function over our dataset $\mathcal{D}$. Formally, $y = \arg \min_{y'} E(y'; \mathcal{D})$. For instance, in linear regression, this error function is

$$E(y'; \mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_{n=1}^{|\mathcal{D}|} (y'(x_n) - t_n)^2.$$

Recall that minimizing this training error is not our ultimate goal; what we really want to achieve is good generalization error. In other words, we want to perform well on test examples $(\mathbf{x}, t) \sim p(\mathbf{x}, t)$ that are not seen in the training set. Mathematically, this means that we are interested in obtaining low generalization loss $\mathbb{E}_{\mathbf{x}, t, \mathcal{D}}[(y(\mathbf{x}; \mathcal{D}) - t)^2]$. We have seen in lecture that this can be decomposed into three terms:

$$\mathbb{E}_{\mathbf{x}, t, \mathcal{D}} [(y(\mathbf{x}; \mathcal{D}) - t)^2] = \underbrace{\mathbb{E}_{\mathbf{x}, \mathcal{D}} [(y(\mathbf{x}; \mathcal{D}) - \bar{y}(\mathbf{x}))^2]}_{\text{variance}} + \underbrace{\mathbb{E}_{\mathbf{x}} [(\bar{y}(\mathbf{x}) - \bar{t}(\mathbf{x}))^2]}_{\text{bias}} + \underbrace{\mathbb{E}_{\mathbf{x}, t} [(\bar{t}(\mathbf{x}) - t)^2]}_{\text{noise}}.$$

In this problem, we shall explore this decomposition in more depth.

Problem 1. In plain ol' linear regression, we previously derived that the predictor

$$y(\mathbf{x}; \mathcal{D}) = \mathbf{x}^\top \underbrace{(\mathbf{X}_\mathcal{D}^\top \mathbf{X}_\mathcal{D})^{-1} \mathbf{X}_\mathcal{D}^\top \mathbf{t}_\mathcal{D}}_{\mathbf{w}_\mathcal{D}}$$

minimizes the mean squared error $E(y, \mathcal{D}) = E_{\text{mse}}(y, \mathcal{D})$. We denote entities that depend on the training set $\mathcal{D}$ with a subscript $\mathcal{D}$ for clarity. Assume $t(\mathbf{x}) = \mathbf{w}_*^\top \mathbf{x} + \epsilon$ where $\epsilon \sim \mathcal{N}(\epsilon|0, 1)$.

1. Explain why $\bar{t}(x) = \mathbf{x}^\top \mathbf{w}_*$.
2. Explain why $\mathbf{t}_\mathcal{D} = \mathbf{X}_\mathcal{D} \mathbf{w}_* + \epsilon$ where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.
3. Using part (2) or otherwise, show that $\bar{y}(\mathbf{x}) = \mathbf{x}^\top \mathbf{w}_*$.
4. Explain why the bias is zero when the assumption holds.

Eigendecomposition of a matrix. Intuitively, we know that adding regularization into the error function should increase the bias and decrease the variance. However, before we consider the scenario, we need some mathematical tools from linear algebra. Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a square matrix with n independent eigenvectors $\{\mathbf{q}_1, \dots, \mathbf{q}_n\}$ and let $\{s_1, \dots, s_n\}$ be its corresponding eigenvalues (i.e. $\mathbf{A}\mathbf{q}_i = s_i\mathbf{q}_i$). Then, $\mathbf{A}$ can be decomposed into

$$\mathbf{A} = \underbrace{\begin{bmatrix} | & & | \\ \mathbf{q}_1 & \cdots & \mathbf{q}_n \\ | & & | \end{bmatrix}}_{\mathbf{Q}} \underbrace{\begin{bmatrix} s_1 & & \\ & \ddots & \\ & & s_n \end{bmatrix}}_{\mathbf{S}} \underbrace{\begin{bmatrix} | & & | \\ \mathbf{q}_1 & \cdots & \mathbf{q}_n \\ | & & | \end{bmatrix}^{-1}}_{\mathbf{Q}^{-1}}.$$

where $\mathbf{Q} = [\mathbf{q}_1 \cdots \mathbf{q}_n]$ and $S = \text{diag}(s_1, \dots, s_n)$. Furthermore, if $\mathbf{A}$ is not singular and has eigenvalues $\{s_1, \dots, s_n\}$, then $\mathbf{A}^{-1}$ has eigenvalues $\{1/s_1, \dots, 1/s_n\}$.

Problem 2. Suppose $\mathbf{A}$ can be decomposed into $\mathbf{Q}\mathbf{S}\mathbf{Q}^{-1}$. Show that $\mathbf{A} + \lambda\mathbf{I}$ can be decomposed into $\mathbf{Q}\mathbf{S}'\mathbf{Q}^{-1}$ where

$$\mathbf{S}' = \begin{bmatrix} s_1 + \lambda & & \\ & \ddots & \\ & & s_n + \lambda \end{bmatrix}.$$

Problem 3. Let us get back to ridge regression. We previously derived that the predictor

$$y(\mathbf{x}; \mathcal{D}) = \mathbf{x}^\top \underbrace{(\mathbf{X}_{\mathcal{D}}^\top \mathbf{X}_{\mathcal{D}} + \lambda \mathbf{I})^{-1} \mathbf{X}_{\mathcal{D}}^\top \mathbf{t}_{\mathcal{D}}}_{\mathbf{w}_{\mathcal{D}}}$$

minimizes the mean squared error with L_2 -regularization $E(y, \mathcal{D}) = E_{\text{mse}}(y, \mathcal{D}) + E_{\text{reg}}(y, \mathcal{D})$. Assume $t(\mathbf{x}) = \mathbf{w}_*^\top \mathbf{x} + \epsilon$ where $\epsilon \sim \mathcal{N}(\epsilon|0, 1)$. Derive the bias term and comment on the effect of the regularization constant λ .